Late Jurassic and Early Cretaceous Radiolarian Age Determinations from Timor-Leste

Munasri[1], Tim R. Charlton[2,3], Maria Guterres[3] and Dino Gandara[3,4]

[1]Sekolah Tinggi Teknologi Mineral Indonesia (STTMI), Bandung, Indonesia
[2]Guildford, U.K.
[3]Timor GAP, E.P., Dili, Timor-Leste.
[4]Sunda Gas Banda Lda., Timor-Leste.

Corresponding author: timrcharlton@gmail.com

Abstract

Two new sets of palaeontological age determinations based on radiolaria are reported from Timor-Leste. The first is from the Suai Loro-1 petroleum exploration well, located near Suai town in SW Timor-Leste, drilled by Timor Oil in 1971. Newly rediscovered ditch cutting samples from the previously undated basal section of this well have yielded radiolarians of Late Jurassic (middle Oxfordian to middle Kimmeridgian) age. On lithology, this stratigraphic interval of reddish shales and interbedded limestones is assigned to the Tchinver Formation.

A second set of radiolarian determinations is from samples collected in the Caraulun river south of Samé town in south-central Timor-Leste. These are dated to the Early Cretaceous (late Valanginian-Hauterivian). The outcrop succession of radiolarian cherts, sandstones and red shales are assigned to the Wai Bua Formation as defined in East Timor (Timor-Leste), which is equivalent to the Nakfunu Formation in Indonesian West Timor.

Manuscript received: 24 Mar 2024, revised manuscript received: 15 Apr 2024, final acceptance: 22 Apr 2024. DOI: 10.51835/bsed.2024.50.1.455

INTRODUCTION

Timor island (Figure 1), located in eastern Indonesia and Timor-Leste, has a varied stratigraphic succession extending back to at least the Permian (Audley-Charles, 1968; Davydov et al., 2014), with previous studies yielding a substantial database of palaeontological age control, although still with much remaining to be learnt from ongoing stratigraphic investigations. Age dating in Timor has employed a wide variety of fossil types, ranging from early studies based on rich Permo-Triassic macrofossil assemblages spectacularly documented, for instance, in the monographic series *Paläontoloie von Timor* (ed. J. Wanner, 1914-1929), to more recent studies employing various microfossil groups including palynology and benthic and planktonic foraminifera (Carter et al., 1976; Haig, 2012; Haig and McCartain, 2007, 2010, 2012; Haig et al. 2018, 2021a and b, etc.). Rather fewer studies of Timor have employed radiolaria (Crespin and Belford, 1959; Harsolumakso et al., 1995; Clowes, 1997; Haig and Bandini, 2013; Munasri and Sashida, 2018; Munasri and Harsolumakso, 2020), although radiolaria can provide a high degree of stratigraphic precision and are particularly useful for dating sedimentary successions originating from relatively deep and/or open marine sedimentary environments. In the case of Timor this applies particularly to what can be considered the principal 'breakup' successions at the transition from Permian-Jurassic relatively shallow marine sedimentary intervals to deeper marine successions that typify the Cretaceous and Tertiary (Charlton, 1989; Sawyer et al., 1993; Figure 2). The present paper outlines two examples from the Late Jurassic and Early

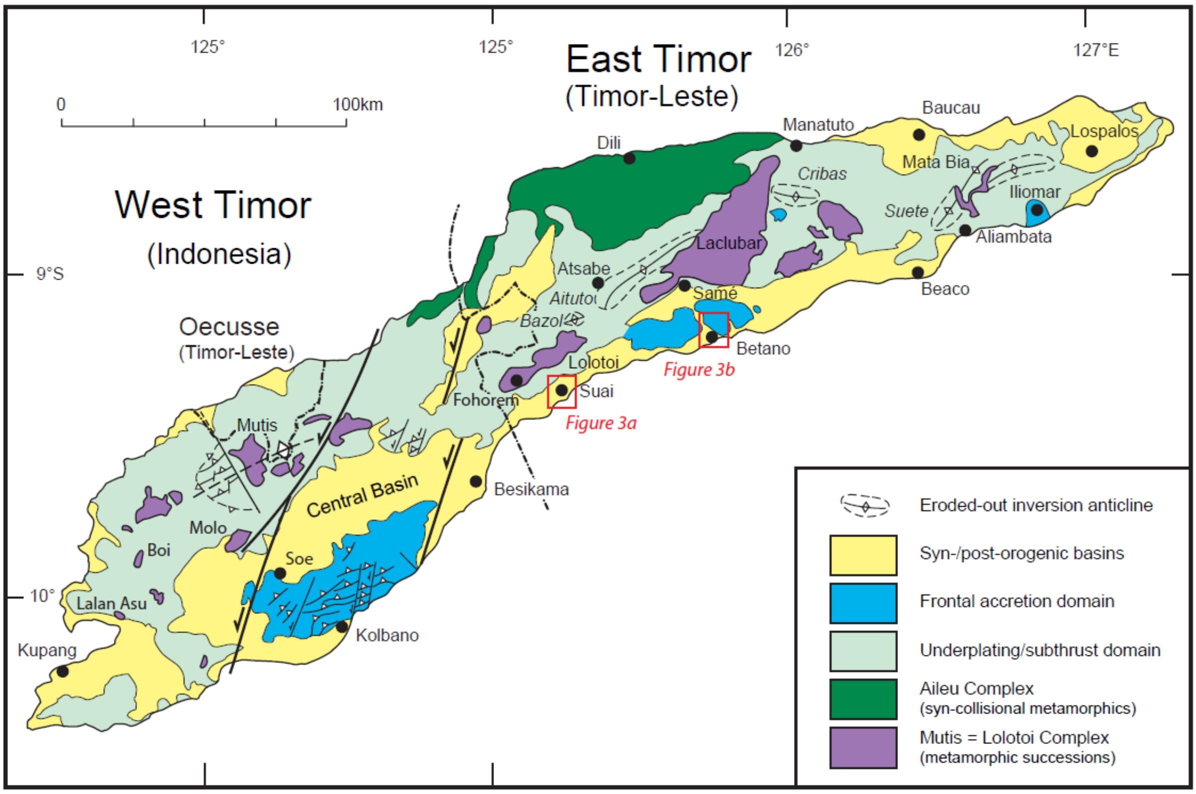

Figure 1: *Timor island tectonostratigraphic domains and location map. The locations of the local geological maps in Figure 3 are indicated.*

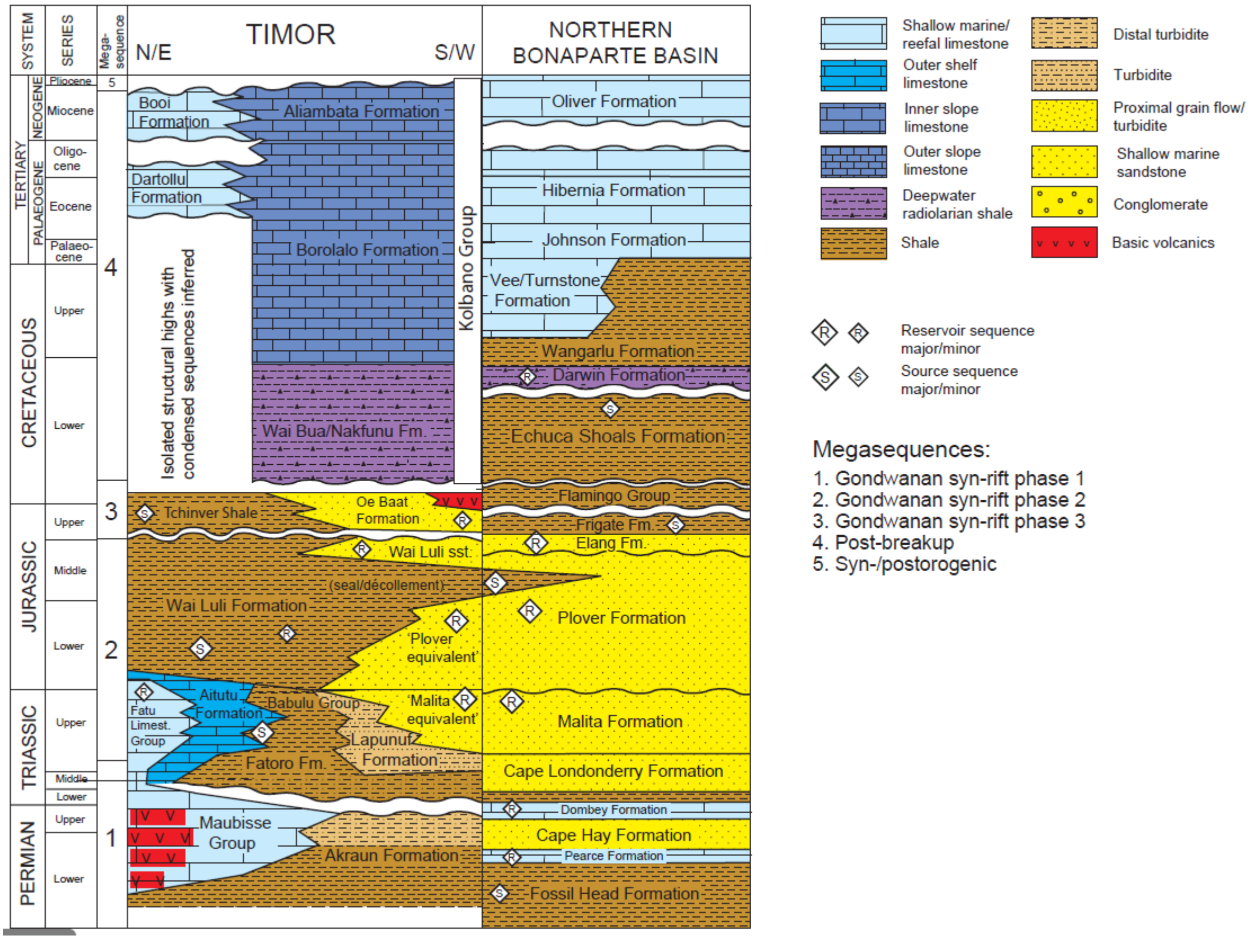

Figure 2: *Timor-Northwest Shelf (Australia) lithostratigraphy, partially updated from Charlton (2002) and Charlton et al. (2020).*

Cretaceous where radiolaria provide precise age controls for stratigraphic successions that were previously rather poorly dated.

Suai Loro-1 well

The Suai Loro-1 petroleum exploration well, located in the southwest of onshore Timor-Leste some 5km to the southeast of Suai town (Figure 3a), was drilled by Timor Oil Ltd. in 1971 (GPS-determined location of the wellhead at 9.33655°S, 125.28570°E). The well penetrated an interval of Neogene synorogenic sediments (Suai and Viqueque Formations) down to a depth of 3305' (1007m), then shale melange assigned to the Bobonaro Complex down to 4990' (1521m; Figure 4). Below this the well drilled an undated interval down to the base of the well at 5088' (1551m) consisting of red shales interbedded with micritic limestones.

The material examined in this study comprised 10 ditch cutting samples from the depth interval 4990–5088', in the lowermost interval of the well section (Table 1). The samples had previously been examined by Robertson Research International who had been unable to date this interval on the basis of palynology (Cater et al., 1971 unpublished report), but had noted the occurrence of radiolaria in the samples, although this information was not acted upon by Timor Oil.

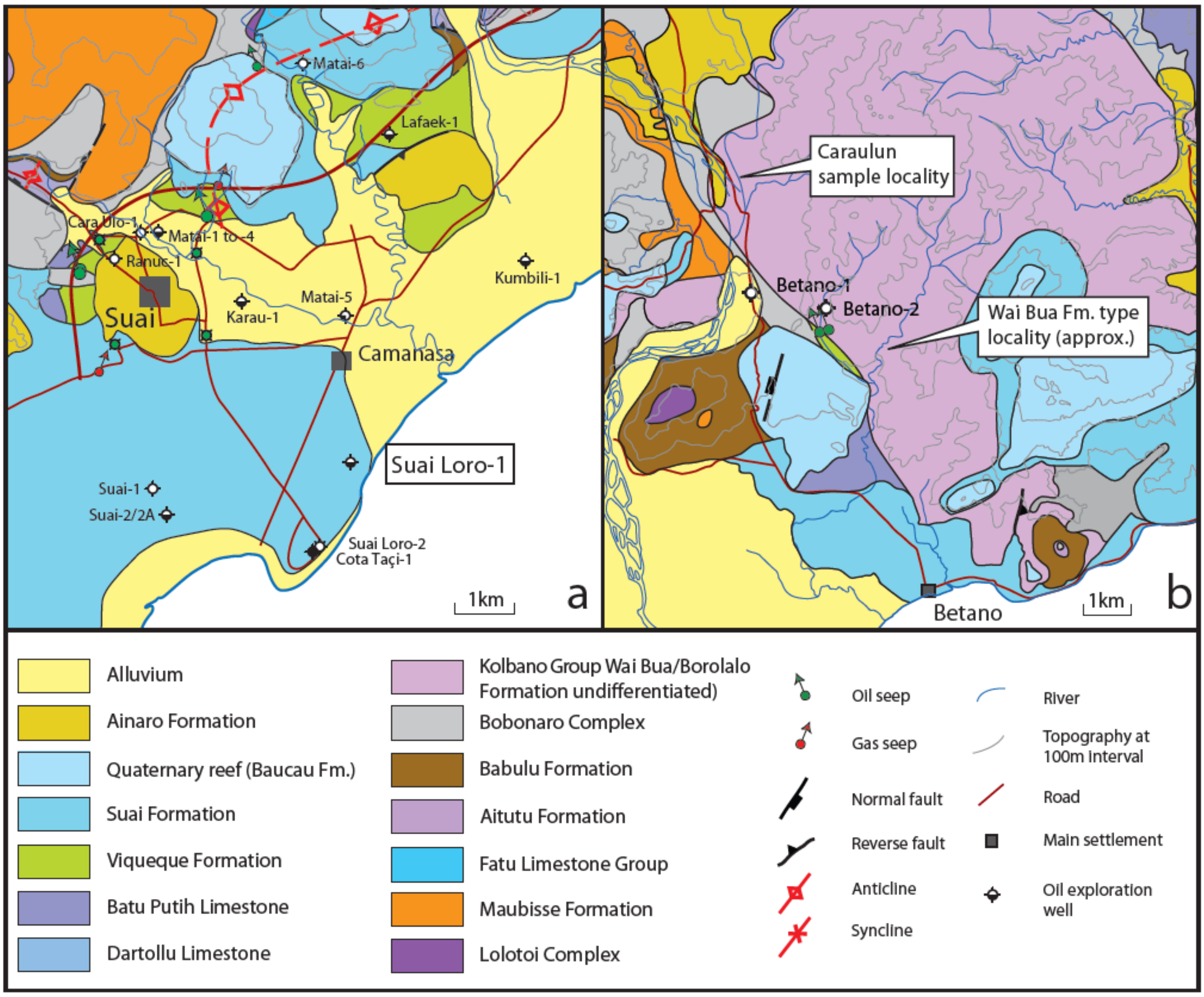

Figure 3: *Geological location maps for the samples reported in this study. (a) Suai area geology, with the location of the Suai Loro-1 well. (b) Betano area geology, showing the location of the Wai Bua Formation outcrop samples analysed in this study, and the approximate location of the Wai Bua type locality proposed by Audley-Charles (1968), although this type's section has not been definitively re-located. The geological mapping in this figure is based on fieldwork undertaken by the Timor GAP Onshore Block geological team (TC, DG, MG and Debora Freitas) in 2017-2019.*

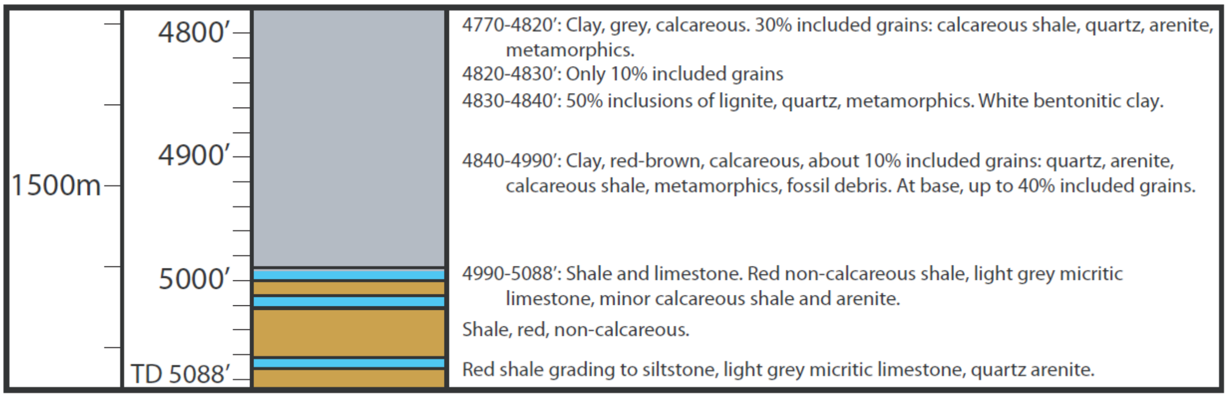

Figure 4: *Lithologies drilled in the basal section of the Suai Loro-1 well. Grey interval is the Bobonaro Complex. Undated shales (brown) and limestones (blue) in the lowest interval drilled were previously undated. Lithological descriptions are from the Timor Oil end of well report (1971).*

In 2019 the ditch cutting samples were relocated in deep storage by Andy Livsey, Technical Director of P.T. Horizon Geoconsulting, in the sample collections from the former Robertson Research (Singapore) office.

Table 1: *List of samples from the interval 4990'–5088' Suai Loro-1 well, Timor Leste.*

No.	Top Depth (feet)	Weight (gr)	Sample Type	Remarks
1-4990	4990	27	Ditch Cuttings	Unwashed, dark gray, argillaceous sand/graywacke, quartz, benthic foraminifera, friable, calcareous.
2-4990		11	Ditch Cuttings	Washed, light-dark brown shale, gray, black and whitish various rock chips, benthic foraminifera and pelecypod.
3-5040	5040	9	Ditch Cuttings	Washed, light-dark brown shale, white–gray limestone, dark color metamorphic? rocks, calcareous.
4-5050	5050	7	Ditch Cuttings	Washed, light-dark brown shale, white–gray limestone, dark color metamorphic? rocks, calcareous.
5-5050		5	Ditch Cuttings	Washed, dark brown shales, gray–white limestone, dark color metamorphic? rocks, calcareous.
6-5060	5060	10	Ditch Cuttings	Washed, light brown shales, white–gray limestone, metamorphic? rocks.
7-5060		3	Ditch Cuttings	Washed, light–dark brown shales and poorly fissile shales, calcareous
8-5070	5070	14	Ditch Cuttings	Washed, light gray–light brown shales, dark color metamorphic? rocks, quartz, benthic foraminifera, calcareous.
9-5070		4	Ditch Cuttings	Washed, light brown shales, limestone, quartz, dark color metamorphic? rocks, quartz, benthic foraminifera, calcareous.
10-5080	5080	11	Ditch Cuttings	Washed, gray shales, limestone, dark color metamorphic? rocks, calcareous.

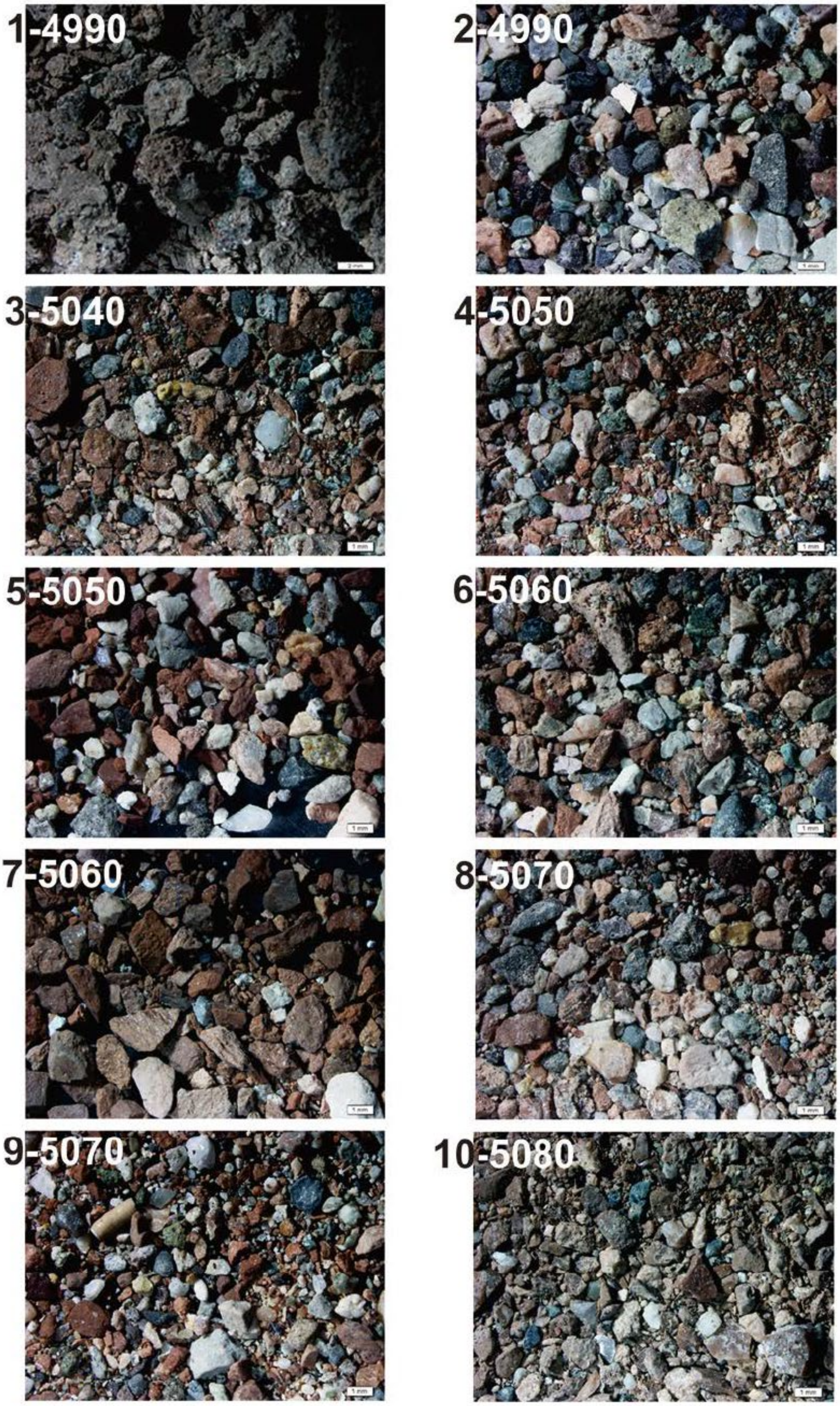

Figure 5: *Photomicrographs of 10 samples from the interval 4990'–5088' in the Suai Loro-1 well, Timor Leste. Note the many mixed lithologies, including likely metamorphic grains. Scale bar in the bottom-right of each photo = 1 mm.*

These samples were made available to Timor GAP for further analysis, with the lead author contracted to re-examine this material.

The ditch cutting samples (Table 1) contain rock grains up to 5mm diameter with sub-angular to sub-rounded grain shapes (Figure 5). The grains consist predominantly of silty shale, light to reddish-brown in colour, and poorly fissile. Subordinate grains in the cuttings include grey to whitish limestone (calcite), dark grey to black metamorphic(?) rock fragments, and quartz. Benthic foraminifera (*Operculina* spp.) were found in samples No. 1-4990, 2-4990, 8-5070, and 9-5070, and a pelecypod-like shell was also found in sample No. 2-4990 (Figure 6). *Operculina* is a genus limited to Late Paleocene-Recent shallow marine deposits (J.T. van Gorsel, reviewer

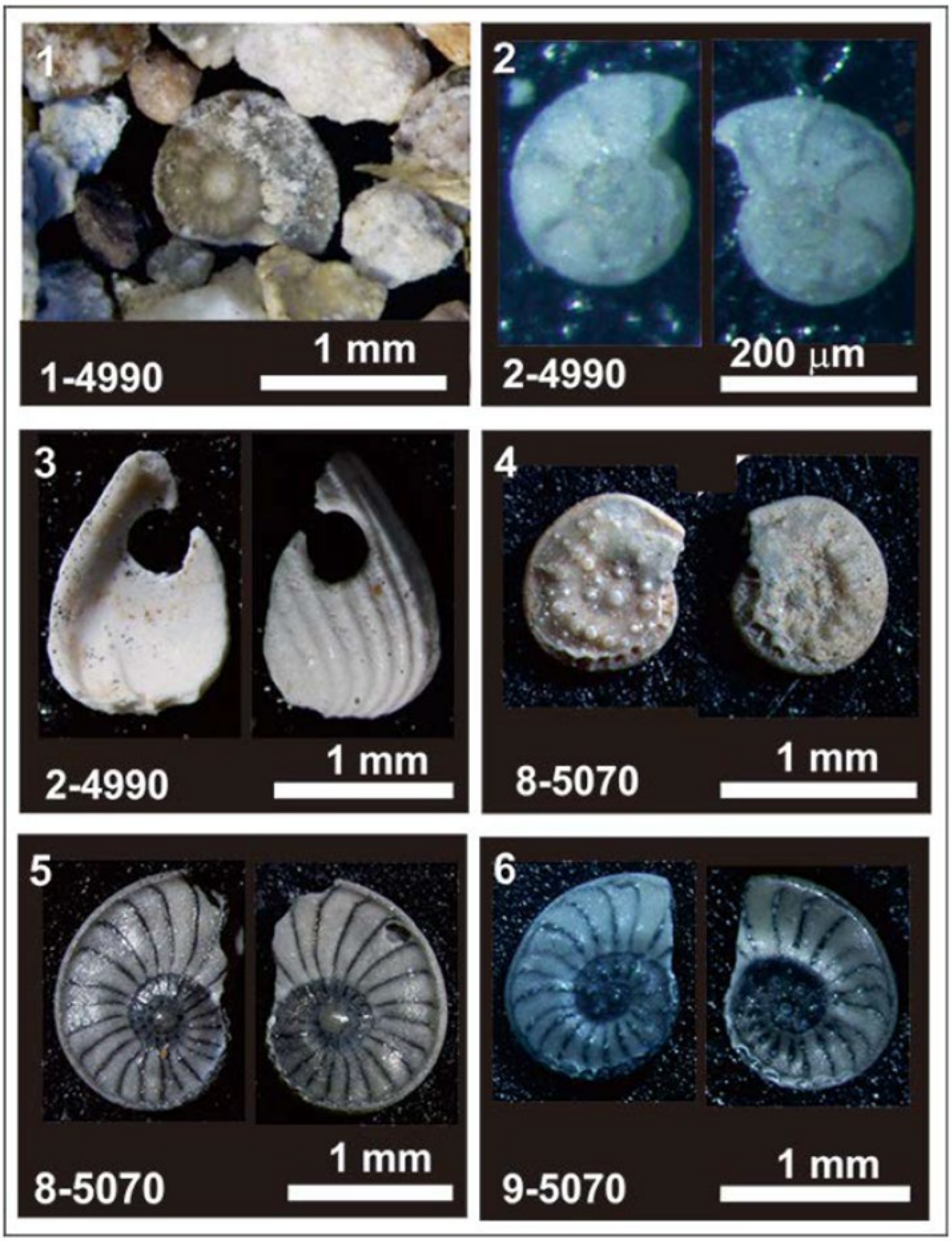

Figure 6: *Photomicrographs of benthic foraminifera and a pelcypod from Suai Loro-1 well. 1–2, 4–6. Operculina spp; 3. Pelecypod. The sample numbers with the top depth are shown in the bottom-left of each photo.*

observation), and these fossils probably derive by caving from the shallower well section.

The ten samples of ditch cuttings were examined closely, but no free radiolaria were observed. However, radiolaria were extracted from shales by placing picked samples in 15% hydrochloric acid until reaction with all calcium carbonate had ceased, then soaked in 3-4% hydrofluoric acid solution for 24 hours. All samples were sieved through 200μm and 63μm nylon screens and then dried. Two of the 10 samples, No. 5-5050 and No. 7-5060, yielded poorly to moderately preserved radiolarians. Radiolarian specimens were then picked and placed in a Scanning Electron Microscope (SEM) holder to be photographed. The faunas of these two samples are listed in Table 2, and photomicrographs of key index species of the radiolarians are shown in Figure 7.

Table 2. *Observed faunas within samples No. 7-5060 and 5-5050 of Suai Loro-1 well*

Radiolarian taxa in sample 7-5060	*Alievium* sp. *Archaeodictyomitra* sp. *Arcanicapsa funatoensis* AITA *Eucyrtidiellum*? sp. *Hiscocapsa* spp. *Hsuum* spp. *Loopus* spp. *Parahsuum* sp. *Parvicingula dhimenaensis* BAUMGARTNER *Protunuma japonicus* MATSUOKA and YAO *Sethocapsa* sp. A. sensu BERTOLINI *Thanarla* sp. cf. *T. patricki* (KOCHER) *Tricolocapsa ruesti* TAN SIN HOK *Williriedellum carpathica* DUMITRICA *Williriedellum yahazuense* (AITA) *Williriedellum* spp. *Xitomitra annibill* (KOCHER) *Xitomitra* sp. cf. *X. tairai* (AITA) *Zhamoidellum ovum* DUMITRICA *Zhamoidellum* spp. *Zhamoidellum ventricosum* DUMITRICA
Radiolarian taxa in sample 5-5050	*Archaeodictyomitra* spp. *Gongylothoax* spp. *Hsuum cuestaensis* PESSAGNO *Loopus primitivus* MATSUOKA *Nassellaria* gen. et sp. indet. *Parahsuum* sp. *Parvicingula* sp. *Sethocapsa* spp. *Sethocapsa* sp. A. sensu BERTOLINI *Syringocapsa*? sp. *Thanarla* sp. cf. *T. patricki* (KOCHER)

	Tricolocapa spp. *Williriedellum* spp. *Williriedellum carpathica* DUMITRICA *Williriedellum yahazuense* (AITA) *Zhamoidellum* spp. *Zhamoidellum ovum* DUMITRICA *Zhamoidellum* sp. cf. *Z. ovum* DUMITRICA.

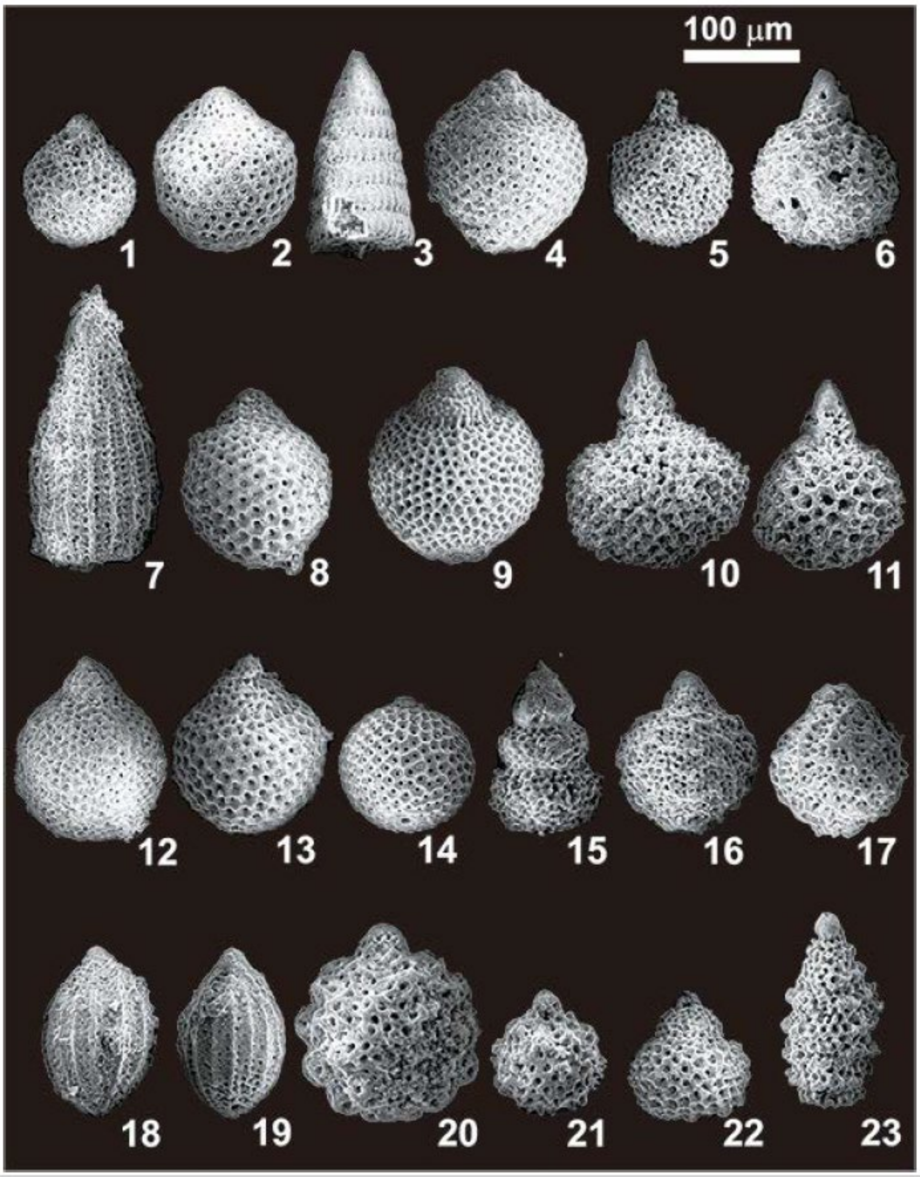

Figure 7: *SEM photomicrographs of Late Jurassic radiolarians from Suai Loro-1 well, Timor Leste. Scale bar = 100 µm. 1–7 are radiolarians from sample 5-5050 and 8–23 are radiolarians from sample 7-5060. 1–2, 8, 12–13. Zhamoidellum ovum; 3. Loopus primitivus; 4, 9. Williriedellum carpathica; 5–6, 10–11. Sethocapsa sp. A. sensu Bertolini; 7. Hsuum cuestaensis; 14. Gongylothorax favosus; 15. Xitomitra annibill; 16–17. Williriedellum yahazuense; 18–19. Protunuma japonicus; 20–21. Arcanicapsa funatoensis; 22. Zhamoidellum ventricosum; 23. Parvicingula dhimenaensis.*

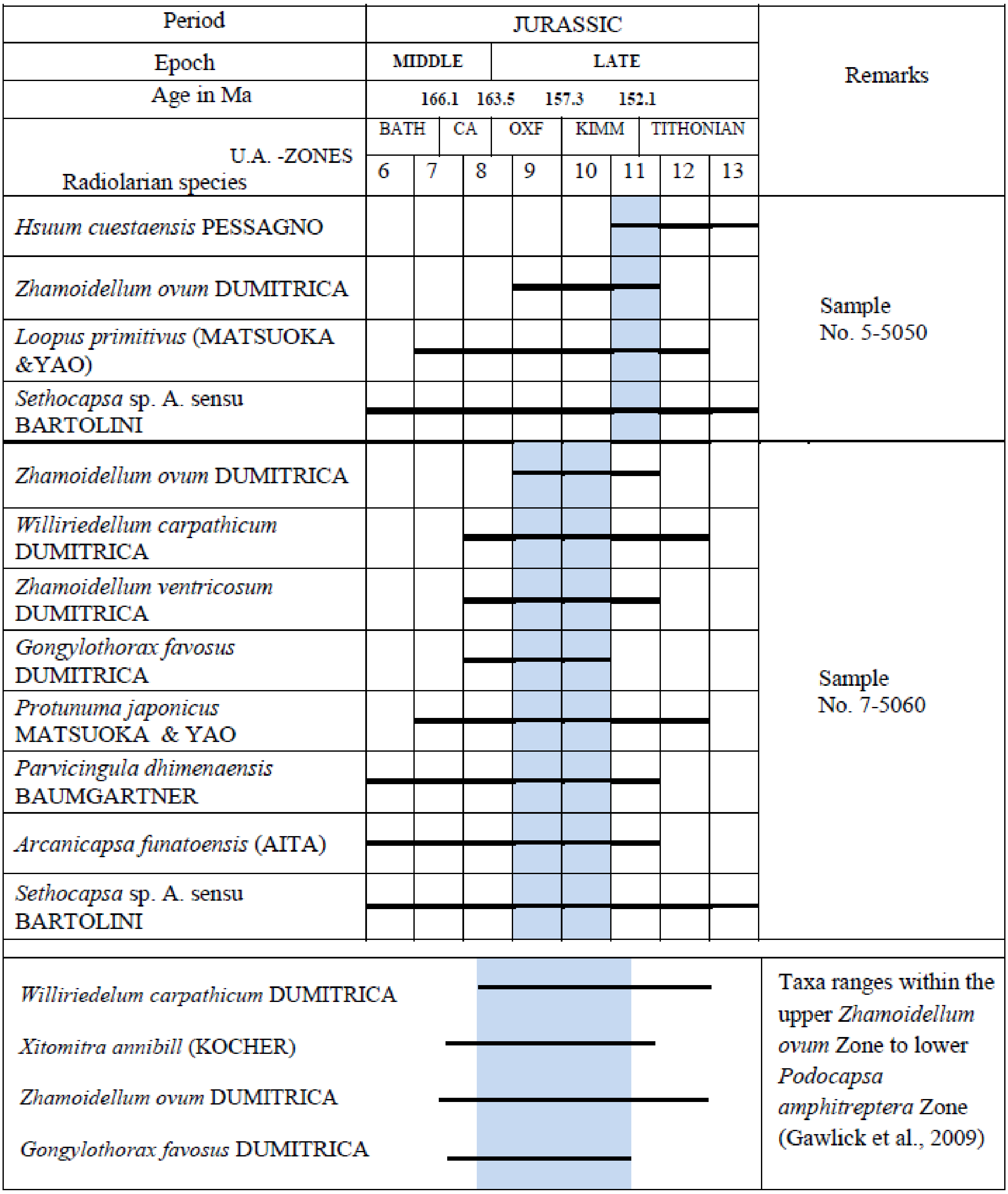

Figure 8: *Biostratigraphic range of selected radiolarian species from samples (No. 7-5060 and No. 5-5050) of the Suai Loro-1 Well, Timor Leste based on the Unitary Association Zones (Baumgartner et al., 1995) and Jurassic radiolarian zonation of Gawlick et al. (2009). Chronostratigraphic age in Ma after International Commission on Stratigraphy (2020).*

Some taxa from Suai Loro-1 have never been described and are probably new. The radiolarians are predominantly cryptocephalic and cryptothoracic nassellarian forms (Dumitrica, 1970), with additionally a few types of multicyrtid nassellarians, and a very few spumellarian types. Sample No. 7-5060 predominantly contains the genus of *Zhamoidellum* including *Zhamoidellum ovum*, while sample No. 5-5050 contains many genera of *Tricolocapsa* and *Sethocapsa* that have not been described before. These are listed in Table 2.

Several taxa are recognized as index fossils and are used for age assignment, including *Arcanicapsa funatoensis*, *Gongylothorax favosus*, *Hsuum cuestaensis*, *Loopus primitivus*, *Parvicingula dhimenaensis*, *Protunuma japonicus*, *Sethocapsa* sp. *A* (sensu Bertolini in Baumgartner et al., 1995), *Xitomitra annibill*, *Williriedellum carpathica*, *Zhamoidellum ovum*, and *Zhamoidellum ventricosum* (Figure 8). Some of the generic names of these radiolarian species were reassigned following O'Dogherty et al. (2017).

To determine a more detailed age based on the radiolarian assemblages, the zonation scheme of Baumgartner et al. (1995: Unitary Association Zones = U.A. Zones) was employed. The association of species in sample No. 7-5060 suggests U.A. Zone 9–10, ranging from middle Oxfordian to late middle Kimmeridgian; while sample No. 5-5050 from the overlying section contains radiolarians indicating U.A. Zone 11 (within the late Kimmeridgian to early Tithonian; Figure 8). By the presence of these radiolarians in the interval represented by the top depth of 5050-5060', it is interpreted that the radiolarian-bearing shales were deposited within the middle Oxfordian to early Tithonian (Late Jurassic) stages, approximately 160–150 Ma according to the International Commission on Stratigraphy Time Scale (2020).

For comparison, the radiolarian biostratigraphic range in the zonation proposed by Suzuki and Gawlick (2003) and modified by Gawlick et al. (2009) is also shown. Among species of radiolarians is an assemblage that belongs to the upper *Zhamoidellum* Zone to lower *Podocapsa amphitreptera* Zone, or within the late Callovian–late Kimmeridgian age range (Figure 8). If we combine the radiolarian age ranges of these two zonations, it suggests that the shales were deposited in the middle Oxfordian to late middle Kimmeridgian interval, approximately 160–154 Ma (International Commission on Stratigraphy Time Scale, 2020). However, it is uncertain yet whether this age range is applicable to the entire stratigraphic interval 4990–5088'.

WAI BUA FORMATION RADIOLARITES IN MOTA (RIVER) CARAULUN

The second set of radiolarian-bearing rocks examined in this study were collected from outcrop 12km SSE of Samé town in the Caraulun river (not to be confused with the river of the same name near Suai town) and 10km NNW of Betano village (Figure 3b). The section sampled outcrops on the left (east) bank of the river between GPS coordinates of 9.08727°S, 125.69468°E and 9.08703°S, 125.69462°E. The field sampling was carried out by TC and MG, following earlier reconnaissance by DG.

The section (Figure 9) exposes reddish to light grey radiolarites alternating with lenses of calcareous and parallel laminated sandstone, with common manganese traces and nodules, and with minor burrowing structures. The rocks are moderately deformed, with fracturing and minor calcite veining. The outcrop section is exposed over a distance of about 30m, dipping moderately (measured dips 37° and 51°) to the S and W.

Figure 9: *Outcrop photos of the Wai Bua Formation Caraulun section, with the sample bags photographed at their position of collection. Sample numbering is TGC-63 (top right) to TGC-70 (bottom right).*

Based on the age determinations (see below), the succession youngs to the south, indicating that the section is right-way-up. This outcrop section is assigned to the Wai Bua Formation, which was defined by Audley-Charles (1968) from a type section reported approximately 4.5km to the SE of the present section, although the original type section has not yet been relocated. Audley-Charles (1968) described the Wai Bua Formation as a succession of brittle radiolarites, radiolarian marls and shales dated as Aptian-Albian (latest Early Cretaceous) based on radiolaria and foraminifera, and younger (Late Cretaceous) foraminiferal limestones (see further discussion of the Wai Bua Formation stratigraphy below).

Eight rock samples were collected, listed in Table 3. In the laboratory the lead author treated the samples in a 5% hydrofluoric acid solution to extract the radiolarian fossils, with 4 samples yielding radiolarian faunas of Valanginian to Hauterivian (Early Cretaceous) age. Lithologically these four samples were described in the laboratory as light gray marl (TGC-63 and TGC-67), light brown calcareous mudstone (TGC-69), and light brown mudstone, non-calcareous or only slightly calcareous (TGC-70). The radiolarian faunas extracted and identified are listed in Table 4.

Table 3: *Sample numbers and field descriptions, Wai Bua Formation, Mota Caraulun-Samé, Timor-Leste.*

Sample #	Location	GPS	Description
TGC 63	Lower Caraulun River	9.08727°S, 125.69468°E	Pinkish to light greyish, medium grained, hard, calcite veined and fairly calcareous.
TGC 64			Brownish, parallel laminated and moderately calcareous claystone with minor manganese traces.
TGC 65			Pinkish, parallel laminated and moderately calcareous siltstone with dark grains and minor manganese traces.
TGC 66			Light greyish shale with minor calcite veins and manganese traces, hard, fractured and very minor calcareous.
TGC 67			Light greyish, calcareous, fractured, and medium grained sandstone with minor dark grains.
TGC 68			Hard, pinkish to brownish radiolarite with dark manganese traces and nodules and minor calcite veins.
TGC 69			Reddish to brownish wet clay with manganese and radiolarite clasts, non-calcareous.
TGC 70		9.08703°S, 125.69462°E	Less soft reddish clay, with dark manganese and minor clasts of reddish radiolarite.

Table 4: *Mota Caraulun radiolarian species list*

TGC-63 (latest Valanginian-late Hauterivian)	*Archaeodictyomitra brouweri* (TAN) *Cecrops septemporata* (PARONA) *Cryptamphorella conara* (FOREMAN) *Cyrtocapsa asseni* TAN *Cyrtocapsa grutterinki* TAN *Cyrtocapsa houwi* TAN *Cyrtocapsa molengraffi* TAN *Cyrtocapsa* spp. (non-Tethyan) *Dictyomitra pseudoscalaris* (TAN) *Eusyringum* spp. (non-Tethyan) *Holocryptocanium barbui* DUMITRICA *Holocryptocanium* spp. (non-Tethyan) *Lithomitra pseudopinguis* TAN *Parvicingula* spp. (non-tethyan) *Pseudodictyomitra lilyae* (TAN) *Sethocapsa nobilis* TAN *Stylocryptocapsa verbeeki* TAN *Stypolarcus laboriosus* TAN *Thanarla pulchra* (SQUINABOL) *Tricolocapsa triangulosa* TAN *Tricolocapsa* sp. *Xitus spicularius* (ALIEV)
TGC-67 (latest Valanginian age)	*Archaeodictyomitra brouweri* (TAN) *Cryptamphorella conara* (FOREMAN) *Cyrtocapsa* spp. (non-Tethyan) *Dictyomitra pseudoscalaris* (TAN) *Hermicriptocapsa capita* TAN *Holocryptocanium barbui* DUMITRICA *Holocryptocanium* spp. (non-Tethyan?) *Parvicingula* spp. (non-Tethyan) *Pseudodictyomitra carpatica* (LOZYNIAK) *Pseudodictyomitra lilyae* (TAN) *Pseudoaulophacus* sp. *Stichocapsa* sp. *Stypolarcus laboriosus* TAN *Thanarla pulchra* (SQUINABOL) *Wrangellium puga* (SCHAAF) *Xitus* sp?
TGC-69 (late Valanginian-early Hauterivian)	*Archaeodictyomitra brouweri* (TAN) *Archaeodicyomitra* sp. *Cryptamphrella conara* (FOREMAN) *Cyrtocapsa* sp. *Dictyomitra pseudoscalaris* (TAN) *Eusyringium* spp. (non-Tethyan) *Holocryptocanium* spp. (non-Tethyan?)

	Lithocampe pseudochrysalis TAN *Parvicingula* spp. (non-tethyan) *Patulella* sp *Praeconocayomma* sp. *Pseudoaulphacus* sp. *Pseudoaulphacus* florealis JUD *Pseudoeucytis* spp. (non-tethyan) *Sethocapsa* sp. *Stichocapsa*sp. *Tricolocapsa* sp.
TGC-70 (late Valanginian)	*Archaeodictyomitra brouweri* (TAN) *Archaeodictyomitra* sp. *Cryptamphorella* sp. *Cryptamphorella macropora* DUMITRICA *Dictyomitra pseudoscalaris* (TAN) *Cyrtocapsa asseni* TAN *Cyrtocapsa molengraaffi* TAN *Cyrtocapsa* sp. *Eusyringium* spp. (non-Tethyan) *Lithocampe pseudochrysalis* TAN *Parvicingula* spp. (non-Tethyan) Parvicingulid spp. (non-Tethyan) *Patulella* sp. *Praeconocaryomma* sp. *Pseudoaulophacus* sp. *Pseudoeucyrtis* spp. (non-tethyan) *Pseudoeucyrtis*(?) sp. *Pseudoeucyrtis hanni* (TAN) *Pseudodictyomitra lilyae* (TAN) *Sethocapsa* sp. *Tricolocapsa* sp.

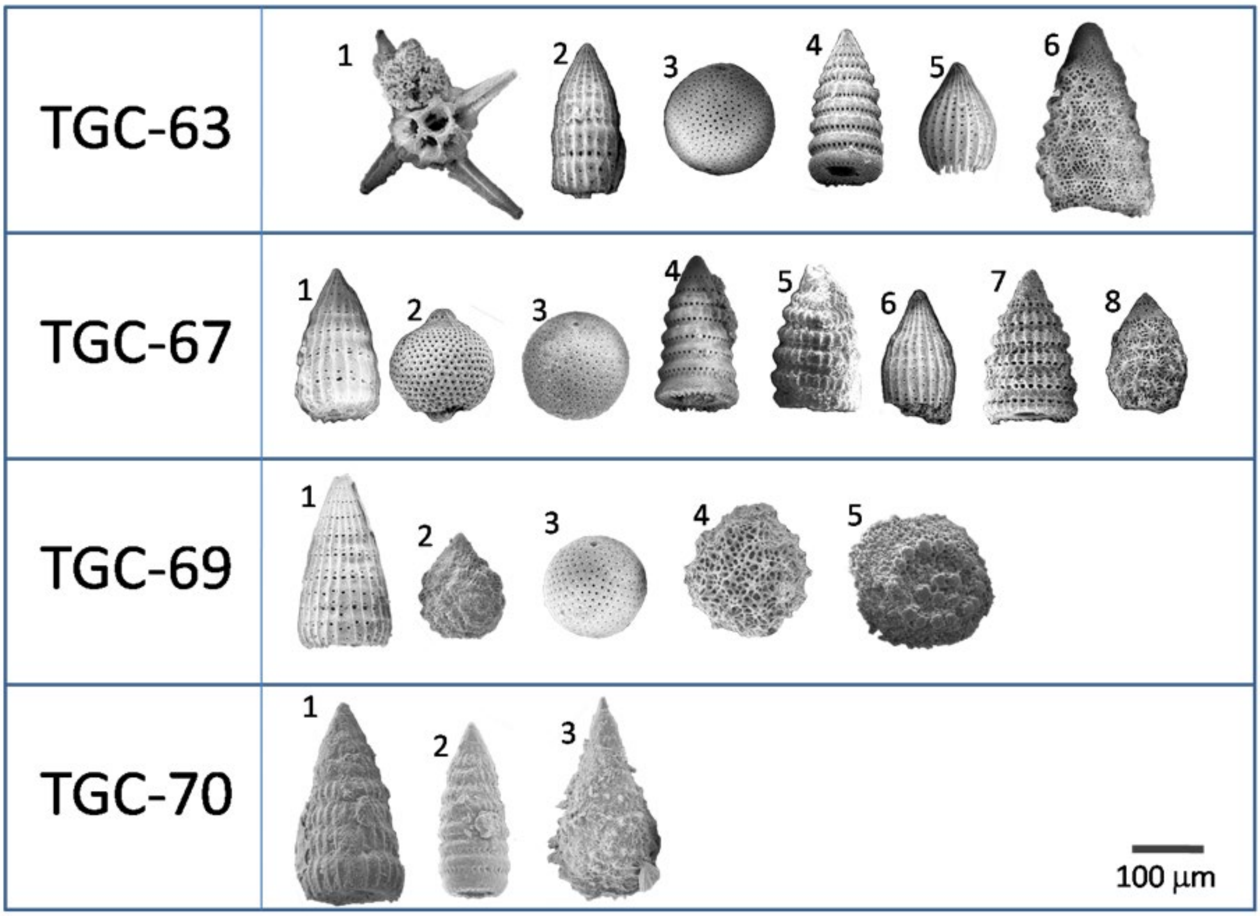

Figure 10: *Radiolarian species used for age determination of the Mota Caraulun samples. Bar scale for all specimens = 100 µm. Sample TGC-63: 1. Cecrops septemporata (PARONA), 2. Dictyomitra pseudoscalaris (TAN), 3. Holocryptocanium barbui DUMITRICA, 4. Pseudodictyomitra lilyae (TAN), 5. Thanarla pulchra (SQUINABOL), 6. Xitus spicularius (ALIEV). Sample TGC-67: 1. Dictyomitra pseudoscalaris (TAN), 2. Hermicriptocapsa capita TAN, 3. Holocryptocanium barbui DUMITRICA, 4. Pseudodictyomitra lilyae (TAN), 5. Pseudodictyomitra carpatica (LOZYNIAK), 6. Thanarla pulchra (SQUINABOL), 7. Wrangellium puga (SCHAAF), 8. Xitus gifuensis MIZUTANI. Sample TGC-69: 1. Dictyomitra pseudoscalaris (TAN), 2. Sethocapsa cetia FORMAN. 3. Holocryptocanium barbui DUMITRICA, 4. Praeconocaryomma prisca PESSAGNO, 5. Pseudoaulophacus florealis JUD. Sample TGC-70: 1. Dictyomitra pseudoscalaris (TAN), 2. Pseudodictyomitra nuda SCHAAF, 3. Pseudoeucyrtis hanni (TAN).*

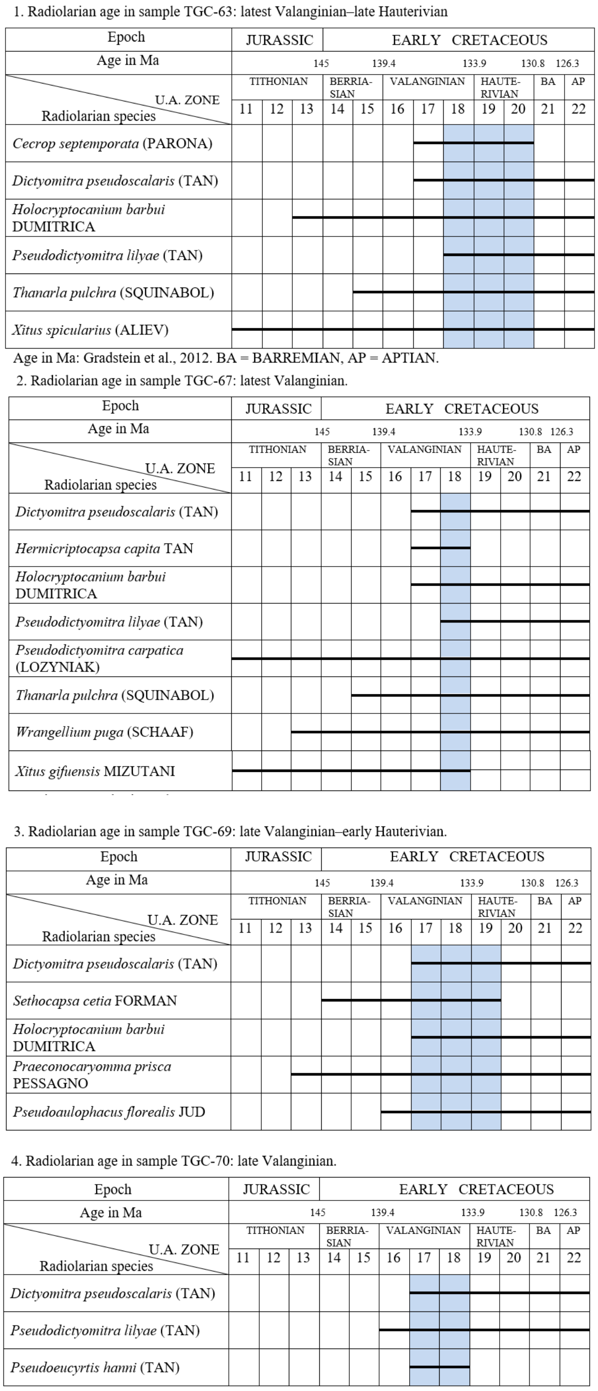

Figure 11: *Zonal ages of samples TGC-63, -67, -69 and -70 based on the Tethyan radiolarian faunal zonation of Baumgartner et al. (1995).*

The faunas contain both Tethyan and non-Tethyan radiolarian faunal elements (in the sense of Baumgartner, 1992, 1993; see also Munasri and Sashida, 2018). The ages of key faunal associations identified (Figure 10) are assigned on the published age zonation of Tethyan type radiolaria, based on the Unitary Association Zone (U.A. Zone) scheme of Baumgartner et al (1995), as indicated in Figure 11.

Sample TGC-70 is dated to the Valanginian, probably the late Valanginian. This is characterized by a relatively greater abundance of non-Tethyan radiolarian types than the other samples, which may indicate an influence of Circum-Antarctic currents or may have developed in geographically higher paleolatitudes. The next sample, TGC-69, is dated to the late or latest Valanginian. It is rather difficult to determine palaeontologically which is older between TGC-70 and TGC-69 because the content of the non-Tethyan radiolaria is fairly equal. TGC-67 is dated to the latest Valanginian, and is the only sample that has a dominance of Tethyan type radiolarians, although several 'non-Tethyan' species are still present. Finally, TGC-63 was deposited during the Hauterivian, containing *Cecrops septemporata* (Pantaneliid form) which has been interpreted as a taxon indicative of low-paleolatitude environments (the Central Tethyan Province: Pessagno and Hull, 2002).

DISCUSSION

Late Jurassic in Suai Loro-1 well

The red shale and limestone succession at the base of the Suai Loro-1 exploration well (Figure 4) is now well dated by this study to the Late Jurassic, and more specifically to the Oxfordian-Kimmeridgian. Similar lithologies of this age have been recorded further east in Timor-Leste as the Tchinver Formation (Charlton and Gandara, 2014; Charlton et al., 2020). The Tchinver river section south of Lospalos (Figure 1) is not yet well dated but consists of green, red and grey shales (although the interpreted Late Jurassic age has been questioned recently by Amaral et al., 2022). Near Aliambata (Figure 1), two samples of grey clay (sample FR-AL-10, 8.79420°S, 126.58067°E; and FR-AL-11, 8.79672°S, 126.57967°E) have been dated to the Oxfordian *Wanaea spectabilis* palynozone (Purcell and Filatoff, 2011 unpublished report). Elsewhere in Timor-Leste Late Jurassic reddish shales were reported regionally by Weber (in Wanner, 1956), while Veevers (1967) recorded Oxfordian red siltstone south of Atsabe, and Collins (2010) mapped mudstones (including a reddish lithofacies) at Lacao near Atsabe, dated to the Oxfordian on nannofossils (determination by J. Backhouse), and also yielding belemnites (*Belemnopsis galoi* of probable Kimmeridgian age) and bivalves (*Malayomaorica malayomaorica* of probable Kimmeridgian-Tithonian age). Near Pualaca Eni (2008) recorded red mudstone containing Late(?) Jurassic belemnites (*Belemnopsis*) and bivalves (*Malayomaorica*) passing upward into grey limestones.

The Tchinver Formation in Timor-Leste is approximately the same age as the Oe Baat Formation in West Timor, which is characterised by glauconitic sandstones, siltstones and shales of probable turbiditic origin (Charlton 1987, 1989; Charlton and Suharsono, 1990; Charlton and Wall, 1994). The glauconite sandstone lithofacies is probably also present at Aliambata in Timor-Leste (location: Figure 1) where glauconitic sandstones were observed (at 8.79475°S, 126.60236°E) stratigraphically above Aitutu Formation (undated here, but regionally Middle-Late Triassic) and grey shales of the Wai Luli Formation (regionally Early Jurassic). We have not yet dated the glauconitic sandstones at Aliambata, but previous authors have reported a *Buchia-Belemnopsis* fauna from this area (e.g. Brunnschweiler, 1977; also Audley-Charles, 1988), a fauna that is also present in the Oe Baat Formation in West Timor. On the other hand, the red-green shales and interbedded limestones of the Tchinver Formation have apparently not yet been recorded in West Timor.

The Late Jurassic radiolarians recovered from the Suai Loro-1 well are comparable but distinct from an equivalent aged fauna previously reported from the Kolbano area of West Timor by Harsolumakso et al. (1995). In their Oxfordian-Tithonian 'Succession E', these authors listed a radiolarian assemblage consisting of *Ristola procera*, *Parvicingula cosmoconica*, *P. dhimenaensis* gr., *Acaeniotyle umbilicata*, *Hsuum* sp., *Stichocapsa* cf. *decora*; and *Pantanellium* sp. and *Acanthocircus trizonalis* (spumellarian type radiolaria). Based particularly on both assemblages containing *Parvicingula*, these successions were probably deposited in the southern Tethyan province, close to the Austral realm (North Austral Province: Pessagno and Hull, 2002, pp. 230, 237).

The Late Jurassic age for the succession at the base of the Suai Loro-1 well may be significant for future petroleum exploration in the Suai area as it suggests the possibility of a coherent and potentially prospective subthrust Triassic-Jurassic succession beneath the Bobonaro Complex in this area.

It is also noteworthy that the Suai Loro-1 ditch cutting samples contain grains of metamorphic rocks. The only metamorphic successions exposed in the Suai area are those of the Lolotoi Complex which was, for instance, intersected at the base of the Cota Taçi-1 well drilled 2km south of Suai Loro-1 (Figure 3a). The Lolotoi Complex is, however, commonly considered an allochthonous tectonostratigraphic element (e.g. Harris, 2006), and should, therefore, have no stratigraphic linkage to the Late Jurassic Tchinver Formation which is clearly part of the Australian margin succession. More detailed petrographic examination of the metamorphic material in the Suai Loro-1 ditch cuttings might prove instructive, but this was beyond the scope of the present study.

The Early Cretaceous Wai Bua Formation

As already mentioned, the stratigraphic section sampled in Mota Caraulun is in an area originally mapped by Audley-Charles (1968) as the Wai Bua Formation. This formation was defined by Audley-Charles as a succession of finely laminated radiolarian shales, multicoloured bedded cherts, brightly coloured brittle radiolarites, biocalcarenites composed essentially of foraminifera and radiolaria with micrite matrix, and calcilutites. Although complexly interthrust, the Wai Bua succession as defined by Audley-Charles clearly consists of two distinct lithological successions, with a lower radiolarian-rich and limestone-poor interval succeeded by a limestone-dominant succession. The lower interval was dated by Audley-Charles to the Aptian-Albian (late Early Cretaceous), with an Albian age determination based on radiolaria (Crespin and Belford, 1959). The limestones were dated to the Late Cretaceous (Cenomanian to Maastrichtian), and re-examination of Audley-Charles's faunal list for these limestones (Audley-Charles, 1968, p.17) employing more recent planktonic foraminiferal faunal zonations (e.g. Boudagher-Fadel, 2013) confirms this age range. Further east at Aliambata and Iliomar (Figure 1), Audley-Charles mapped other Late Cretaceous deepwater limestones as the Borolalo Limestone (Formation), and the faunal list for the Late Cretaceous planktonic foraminifera in the Wai Bua Formation north of Betano overlaps strongly with that of the Borolalo Limestone (Audley-Charles, 1968, faunal list p.18). It seems more appropriate on lithology and age ranges to distinguish the essentially non-calcareous radiolarian shales and radiolarites in the Betano area as one lithostratigraphic formation, and the limestones at Betano, Aliambata and Iliomar as a separate formation. Charlton et al. (2020) have proposed modified definitions of the Wai Bua Formation and Borolalo Formation, with the name Wai Bua Formation restricted to the essentially non-carbonate units of Early Cretaceous age, while the succeeding carbonate-dominated successions, ranging in age from the Aptian-Albian (late Early Cretaceous) and continuing into the Late Cretaceous and Paleogene, are assigned to a modified Borolalo Formation (Figure 12). The Wai Bua and Borolalo Formation are both assigned to the Kolbano Group (Figure 2), defined from West Timor for deepwater sediments of Cretaceous, Paleogene and Miocene age, and widely interpreted as representing the distal edge of the NW Australian continental margin succession (Audley-Charles, 1988; Charlton, 1989; Haig et al., 2018).

The boundary between the top of the (redefined) Wai Bua Formation and the succeeding Borolalo Formation in Timor-Leste appears to be gradational, with transitional intervals developed through the Aptian and Albian stages. In a 1973 unpublished report by D.J. Carter, consultant micropalaeontologist to Timor Oil (the former petroleum concessionaires in East Timor), samples of radiolarian shales and radiolarite assignable to the Wai Bua Formation were dated on planktonic foraminifera to the Aptian-early Albian (2 samples) and early Albian (3 samples), with a sixth sample dated less precisely to the Albian. Five further samples that appear to be from a Wai Bua-Borolalo transitional facies (mixed calcilutite and radiolarite) also yielded early Albian ages (Carter, unpublished reports, 1972a, b), while two similarly transitional lithologies reported by Haig and McCartain (2007, samples UWA134914 and UWA134983) yielded Aptian-early Albian ages. More limestone-dominant samples assignable to the (re-defined) Borolalo Formation include a late Aptian pale red planktonic foraminiferal limestone (Haig and McCartain, 2007), a probably late Aptian-early Albian planktonic foraminiferal wackestone (Benincasa, 2015), and samples from the Betano-2 exploration well (Carter, unpublished report 1972b) dating to the

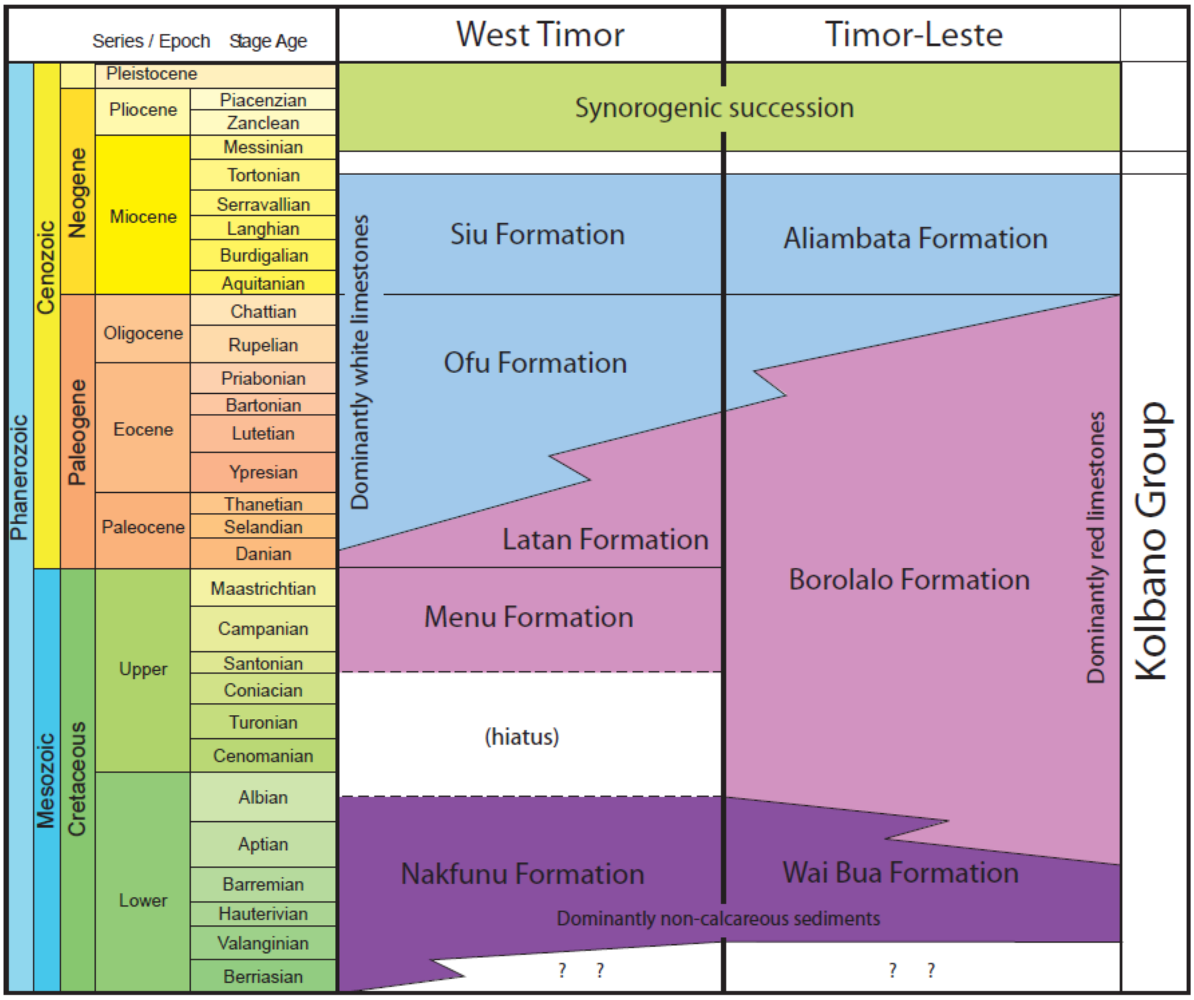

Figure 12: *Timor (West Timor and Timor-Leste) stratigraphic nomenclature for the Kolbano Group (Audley-Charles, 1968; Charlton and Wall, 1994; Sawyer et al., 1993; Clowes, 1997; Munasri and Sashida, 2018).*

early Albian (11 samples), probably early Albian (2 samples), late Albian (1 sample) and undifferentiated Albian (6 samples). Late Cretaceous age determinations for the re-defined Borolalo Formation include the Cenomanian (Carter, 1973), Turonian (Haig and McCartain, 2007; Benincasa, 2015), and Santonian, Campanian, and Maastrichtian (Haig and McCartain, 2007).

Our new late Valanginian-Hauterivian age determinations from the Mota Caraulun section extend the age of the Wai Bua Formation in Timor-Leste down from the previously recognised Aptian-Albian stages (Audley-Charles, 1968) into the older Early Cretaceous (previously suspected, but not definitively established, e.g. Carter, 1973). The minimum age range for the Wai Bua Formation in Timor-Leste is now established as late Valanginian-Albian, which is precisely the same as the age range established for the lithologically equivalent Nakfunu Formation in West Timor (Clowes, 1997), although more recently Munasri and Sashida (2018) have also recognized Berriasian-early Valanginian radiolarian ages (in addition to younger faunas) in the Nakfunu Formation of West Timor. In West Timor Clowes (1997) found no planktonic foraminiferal faunas from the early Late Cretaceous interval (Cenomanian-Coniacian), and she suspected a sedimentary hiatus through this period in West Timor (Figure 12). It remains to be seen whether future sampling will fill this apparent age gap in West Timor.

In terms of faunal content, the Wai Bua assemblage from Mota Caraulun is similar to, but not identical with the fauna of the Nakfunu Formation in West Timor (Munasri and Sashida, 2018) and other locations around the Indian Ocean (Baumgartner, 1992, 1993). It seems that the Mota Caraulun radiolarian assemblage is closer to that from Rote Island (Tan Sin Hok, 1927), based on a relative rarity of spumellarian type radiolarian forms. It may be significant that the oldest of the four dated samples (TGC-70) contains a fauna that suggests the highest southerly palaeolatitude (or at least the greatest southern polar influence), while the youngest sample (TGC-63) indicates more low-latitude conditions, consistent with generally northward movement of the Australian continental margin at this time (cf. Charlton, 2023, figures 9 and 10). A younger (Cenomanian-Turonian; early Late Cretaceous) radiolarian fauna in the Noni Formation of West Timor (Munasri and Harsolumakso, 2020) shows the same low-latitude palaeoenvironment (the Central Tethyan Province of Pessagno and Hull, 2002) as the youngest of the Caraulun samples, although the Noni Formation is widely considered to be an allochthonous structural element (e.g. Harris, 2006), distinct from the predominantly Australian margin 'parautochthonous' stratigraphic successions such as the Wai Bua Formation in Timor-Leste and the directly equivalent Nakfunu Formation in West Timor.

ACKNOWLEDGMENTS

The authors are grateful to the management of Timor GAP, E.P., and particularly Norberta da Costa Noronha, former Managing Director of Timor GAP Onshore Block Lda., for permission to publish this work. Thanks also to Andy Livsey (P.T. Horizon Geoconsulting, Jakarta) for relocating and sharing the Suai Loro-1 well material. The manuscript benefitted from the constructive

comments of J.T. (Han) van Gorsel and two anonymous reviewers.

References

Amaral, A.P., Cardoso, P., Pires, J., Cabral, E. da S., Hornai, A.M. and Ribeiro, G. da C. 2022. Reassessment of the Tchinver Shale type section. *Proceedings of the 5th IPG International Conference on Geoscience Data and Information for Investment and Sustainability Development of Timor-Leste*. IPG, 67-71.

Audley-Charles, M.G. 1968. The Geology of Portuguese Timor. *Mem. Geol. Soc. Lond.* 4.

Audley-Charles, M.G. 1988. Evolution of the southern margin of Tethys (north Australian region) from early Permian to late Cretaceous. In: *Gondwana and Tethys* eds. M.G. Audley-Charles and A. Hallam. *Geol. Soc. Lond. Spec. Publ.* 37, 79-100.

Baumgartner, P. O. 1992. Lower Cretaceous radiolarian biostratigraphy and biogeography off northwestern Australia (ODP Sites 765 and 766 and DSDP Site 261), Argo abyssal plain and lower Exmouth Plateau. In Gradstein, F. M. et al. (eds.), Proceedings of the Ocean Drilling Program, Scientific Results-College Station, TX, 123, 299-343.

Baumgartner, P. O. 1993. Early Cretaceous radiolarians of the Northeast Indian Ocean (Leg 123: Sites 765, 766 and DSDP Site 261): The Antarctic-Tethys connection. In Lazarus, D. and De Wever, P. eds., *Marine Micropaleontology*, 21, 329-352.

Baumgartner, P. O., Bartolini, A., Carter, E. S, Conti, M., Cortese, G., Danelian, T., De Wever, P., Dumitrica-Jud, R., Gorican, S., Guex, J., Hull, D. M., Kito, N., Marcucci, M., Matsuoka, A., Murchey, B., O'Dogherty, L., Savary, J., Vishnevskaya, V., Widz, D. and Yao, A. 1995. Middle Jurassic to Early Cretaceous radiolarian biochronology of Tethys based on Unitary Associations. *Mémoires de Géologie* (Lausanne) 23, 1013-1048.

Benincasa, A. 2015. *The 'fatus' of East Timor: stratigraphy and structure*. PhD thesis, University of Western Australia.

Boudagher-Fadel, M.K. 2013. *Biostratigraphic And Geological Significance Of Planktonic Foraminifera.* 2nd Edition. OVPR UCL, London.

Brunnschweiler, R.O. 1978. Notes on the geology of eastern Timor. *BMR Bull. Aust. Geol. Geophys.* 192, 9-18.

Carter, D.J. 1972a. Report on surface samples form the Betano area. Timor Oil, Palaeontological Report 18, (Unpublished).

Carter, D.J. 1972b. Report on D. J. Warwick's samples from the Betano region. Timor Oil, Palaeontological Report 19, (Unpublished).

Carter, D.J. 1973. Report on Borehole Betano No. 2. Timor Oil, Palaeontological Report 20, (Uunpublished).

Carter, D.J., Audley-Charles, M.G. and Barber, A.J. 1976. Stratigraphical analysis of an island arc-continental margin collision zone in eastern Indonesia. *J. Geol. Soc. Lond.* 132, 179-198.

Cater, M. C., Scrutton, M. E,. Tidey, G. L. and Attewell, R. A. K., 1971. The micropaleontology and stratigraphy of the Timor Oil Suai Loro No. 1 well, Portuguese Timor. Robertson Research International

Limited Singapore Branch, Oilfields Report 47, 18pp., (Unpublished).

Charlton, T.R. 1987. *The Tectonic Evolution of the Kolbano-Timor Trough Accretionary Complex, Timor, Indonesia.* Unpublished Ph.D. thesis, London University.

Charlton, T.R. 1989. Stratigraphic correlation across an arc-continent collision zone: Timor and the Australian Northwest Shelf. *Aust. J. Earth Sci.* 36, 263-274.

Charlton, T.R. 2002. The petroleum potential of East Timor. *APPEA Proc.* 32, 20-38.

Charlton, T.R. 2023. Jurassic-Paleogene spreading history of the northern Indian Ocean as a constraint on the evolution of the north Australian continental margin. *Berita Sedimentologi* 49(2), 75-107.

Charlton, T.R. and Gandara, D. 2014. The petroleum potential of Timor-Leste. Proc. 38th Annual Conv. Indonesian Petroleum Assoc. (IPA), Jakarta, IPA 14-G-017, 7pp.

Charlton, T., Gandara, D., Guterres, M., Freitas, D. and da Costa Noronha, N. 2020. TIMOR GAP's lithostratigraphic framework for Timor-Leste. (Unpublished, available online at https://www.researchgate.net/publication/374386686_Lithostratigraphic_framework_for_Timor-Leste).

Charlton, T.R. and Suharsono, 1990. Mesozoic-Tertiary stratigraphy of the Kolbano area, southern West Timor. *Bull. GRDC* 14, 38-58.

Charlton, T.R. and Wall, D. 1994. New biostratigraphic results from the Kolbano area, southern West Timor: implications for the Mesozoic-Tertiary stratigraphy of Timor. *J. SE Asian Earth Sci.* 9(1/2), 113-122.

Clowes, E. 1997. *Micropalaeontological analysis of the Kolbano Sequence (Jurassic to Pliocene), West Timor, and its radiolarian fauna.* Ph.D. thesis, London University, 443p.

Collins, M.A. 2010. *Stratigraphic reconstruction of a thrust stack succession in the Lacao region, Timor Leste.* Honours thesis, University of Western Australia.

Crespin, I. and Belford, D.J. 1959. Micropalaeontology of further rock samples from Portuguese Timor. *BMR Rec.* 1959/118, 3pp.

Davydov, V.I., Haig, D.W., McCartain, E.W. 2014. Latest Carboniferous (late Gzhelian) fusulinids from Timor Leste and their paleobiogeographic affinities. *J. Paleont.* 88(3), 588-605.

Dumitrica, P., 1970. Cryptocephalic and Cryptothoracic Nassellaria in some Mesozoic deposits of Romania. *Revue Roumaine de Géologie, Géophysique et Géographie (série Géologie)*, Bucharest 14, 1-124.

Eni (P. Giaj-Via and S. Poynter). 2008. *Timor-Leste Field Trip*, unpublished company report.

Gawlick, H.-J., Missoni, S., Schlagintweit, F., Suzuki, H., Frisch, W., Krystyn, L., Blau, J., and Lein, R. 2009. Jurassic tectonostratigraphy of the Austroalpine Domain. *Journal of Alpine Geology* 50, 1–152.

Haig, D.W. 2012. Palaeobathymetric gradients across Timor during 5.7–3.3 Ma (latest Miocene–Pliocene) and implications for collision uplift. *Palaeogeog.*

Palaeoclimatol. Palaeoecol. 331-332, 50-59.

Haig, D.W. and Bandini, A.N. 2013. Middle Jurassic Radiolaria from a siliceous argillite block in a structural melange zone near Viqueque, Timor-Leste: palaeogeographic implications. *J. Asian Earth Sci.* 75, 71-81.

Haig, D.W. and McCartain, E. 2007. Carbonate pelagites in the post-Gondwana succession (Cretaceous-Neogene) of East Timor. *Australian J. Earth Sci.* 54, 875-897.

Haig, D.W., McCartain, E. 2010. Triassic organic-cemented siliceous agglutinated foraminifera from Timor-Leste: conservative development in shallow-marine environments. *J. Foram. Res.* 40(4), 366-392.

Haig, D.W., and McCartain, E. 2012. Intraspecific variation in Triassic Ophthalmidiid Foraminifera from Timor. *Rev. Micropaleontologie*, 55, 39-52.

Haig, D.W., Mossadegh, Z.K., Parker, J.H. and Keep, M. 2018. Middle Eocene neritic limestone in the type locality of the volcanic Barique Formation, Timor-Leste: microfacies and tectonostratigraphic affinities. *J. Asian Earth Sci.*

Haig, D.W., Rigaud, S., McCartain, E., Martini, R., Barros, I.S., Brisbout, L., Soares, J. and Nano, J. 2021a. Upper Triassic carbonate-platform facies, Timor-Leste: foraminiferal indices and regional tectonostratigraphic association. *Palaeogeog. Palaeoclimatol. Palaeoecol.* 570, 22pp.

Haig, D.W., Rigaud, S., McCartain, E., Nano, J., Barros, I.S. and Martini, R. 2021b. Biostratigraphic indices for Lower Jurassic carbonate-platform deposits (Perdido Group), Overthrust Terrane Association, Timor-Leste. J. Asian Earth Sci. 215, 14pp.

Harris, R. 2006. Rise and fall of the Eastern Great Indonesian arc recorded by the assembly, dispersion and accretion of the Banda Terrane, Timor. *Gondwana Research* 10, 207-231.

Harsolumakso, A.H., Villeneuve, M., Cornée, J.-J., De Wever, P., Tronchetti, G., Butterlin, J., Glaçon, G. and Saint-Marc, P., 1995. Stratigraphie des séries paraautochtones du Sud de Timor occidental (Indonésie). Comptes Rendus de l'Académie des Sciences, Series 11A (320), p. 881–888.

Munasri and Harsolumakso, A. H., 2020. Late Cretaceous Radiolarians from the Noni Formation, West Timor, Indonesia. *Berita Sedimentologi* 45, 5-18.

Munasri and Sashida, K. 2018. Tethyan and non-Tethyan Early Cretaceous radiolarian faunas from West Timor, Indonesia: palaeogeographic and tectonic significance. *Earth Evolution Sci.* 12, 3-12.

O'Dogherty, L., Gorican, S. and Gawlick, H.-J. 2017. Middle and Late Jurassic radiolarians from the Neotethys suture in the Eastern Alps. *Journal of Paleontology* 91(1), p. 1–48.

Pessagno Jr., E. A. and Hull, D. M. 2002. Upper Jurassic (Oxfordian) radiolaria from the Sula Islands (East Indies): Their taxonomic, biostratigraphic, chronostratigraphic, and paleogeographic significance. *Micropaleontology* 48(3), 229-256.

Purcell, R. and Filatoff, J. 2011. Palynology report, outcrop samples, East Timor. Unpublished consultancy report.

Sawyer, R.K., Kartono Sani and Brown, S. 1993. Stratigraphy and sedimentology of West Timor, Indonesia. *Proc. 22nd Annual Conv. Indonesian Petroleum Assoc. (IPA), Jakarta,* 1, 533-574.

Suzuki, H. and Gawlick, H.-J. 2003. Die jurassischen Radiolarien zonen der Nordlichen Kalkalpen. In: Weidinger, J. T., Lobitzer, H. and Spitzbart, I. (eds): *Beitrage zur Geologie des Salzkammerguts. Gmundner Geo-Studien*, 2, 115–122, Gmunden.

Tan Sin Hok, 1927. Over de samenstelling en het onstaan van krijten mergel-gesteenten van de Molukken. Jaarboek van het Mijnwezen in Nederlandsch Oost-Indie, 55, (1926), Verhandelingen, 5-165.

Veevers, J.J. 1967. Report on an official visit to Portuguese Timor from 25th April to 9th May, 1967. *BMR Records, Canberra* 1967/4, 8pp.

Wanner, J. (ed.) 1914-1929. *Paläontologie von Timor* (16 volumes). Stuttgart.

Wanner, J. 1956. Zur stratigraphie von Portuguesisch Timor. *Zeitschr. Deutsche Geol. Gesellsch.* 108, 109-140.